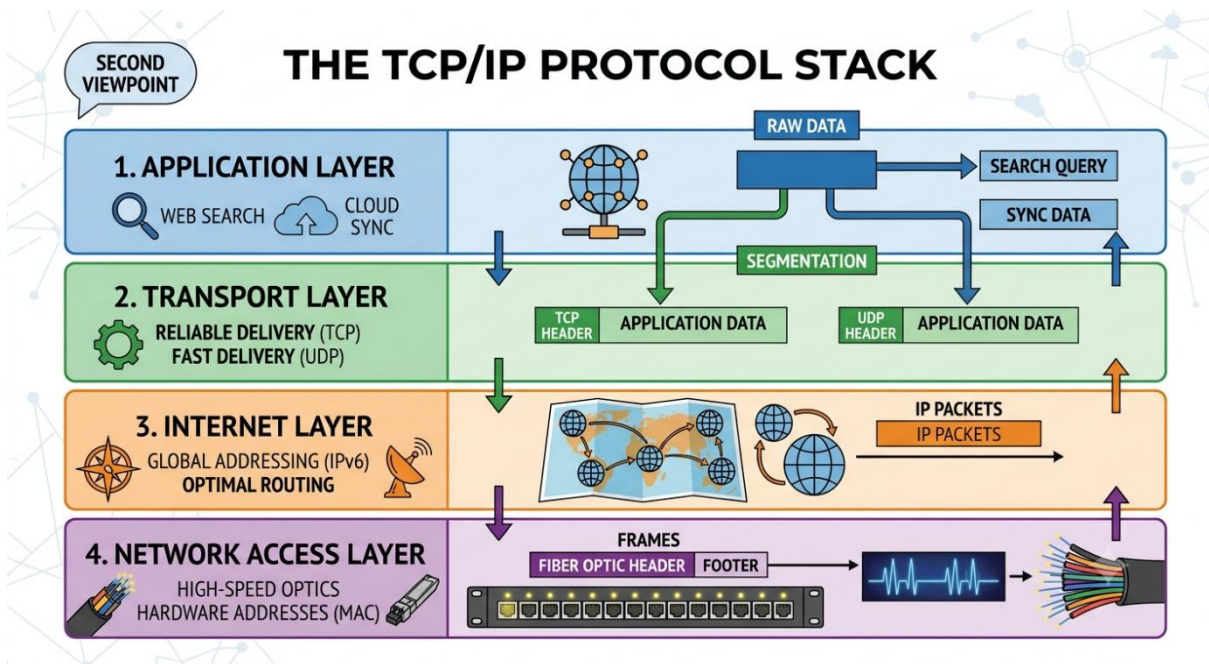


How the Internet Delivers a Web Page

You can use **YouTube** as the example website.

1. TCP/IP Protocol Stack



- Application Layer: Provides services to users. HTTP requests webpages and DNS converts domain names into IP addresses.
- Transport Layer: TCP ensures reliable communication between the browser and server.
- Internet Layer: Uses IP addresses to route packets across networks.
- Network Access Layer: Handles the physical transmission of data through Wi-Fi, Ethernet, etc.

2. Role of DNS

When the user types **www.youtube.com**, the browser does not know the server's IP address.

- Application passes the name to the DNS Resolver.
- Resolver sends a query to the local DNS Server.
- DNS Server returns the corresponding IP address.
- Resolver returns the IP address to the application.

After obtaining the IP address, the browser can communicate with the YouTube server.

3. HTTP Request and Response

HTTP is the protocol used to transfer webpages.

1. Browser creates an HTTP request.
2. The request is sent to the YouTube web server.
3. The server processes the request.
4. The server sends an HTTP response containing the webpage data.
5. The browser displays the webpage to the user.

4. How TCP Provides Reliability

TCP provides reliable communication by:

- Establishing a connection using a three-way handshake.
- Dividing data into segments.
- Numbering each segment.
- Detecting lost packets.
- Retransmitting missing packets.
- Using acknowledgements (ACKs) to confirm delivery.
- Delivering data in the correct order.

Because of TCP, HTTP data reaches the browser accurately and completely.

5. Client–Server or Peer-to-Peer

- Client-Server: One server provides service to multiple clients
Server centrally controls communication
- Peer-to-Peer: Each host can provide services Each host can request services No dedicated server required Peers communicate: Directly with each other

Here YouTube follows a **client–server architecture**.

Justification:

- The browser acts as the **client**.
- YouTube servers store videos and webpages.
- Clients request services, and servers respond.
- Communication is controlled by centralized servers rather than equal peers.

6. HTTP vs FTP

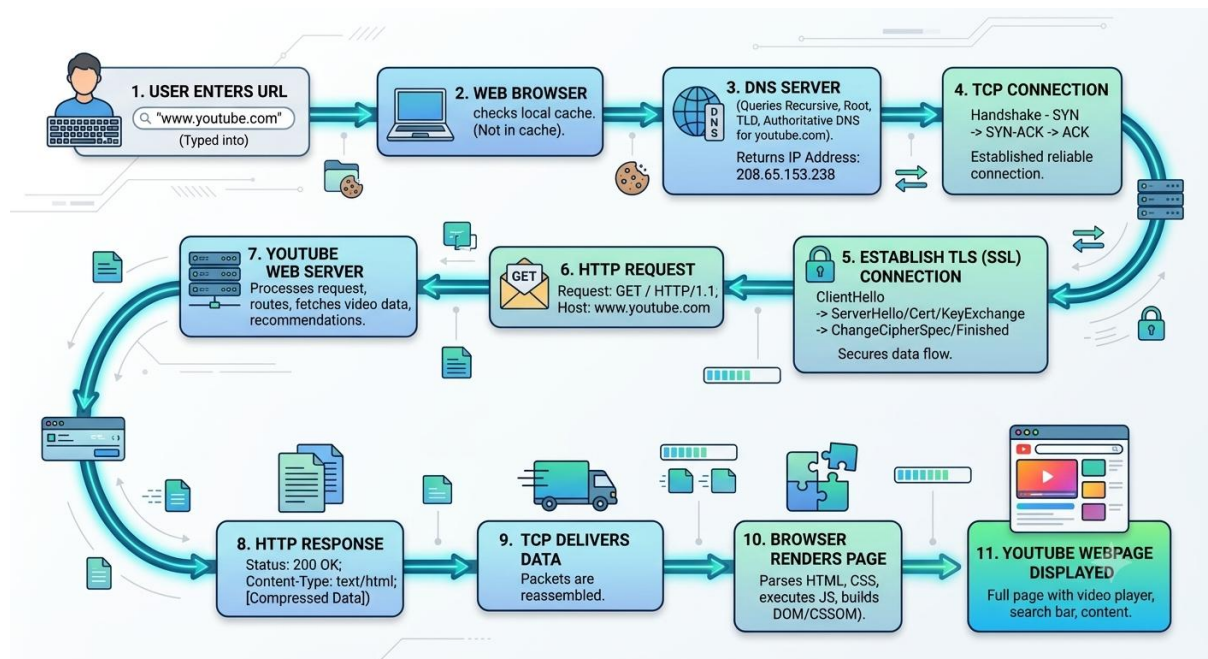
HTTP	FTP
Used to transfer web pages	Used to transfer files
Default port is 80	Default ports are 20 and 21
Transfers HTML, CSS, images, and web resources	Uploads and downloads files
Mainly used by web browsers	Mainly by used FTP clients

7. Cookies and User Experience

Cookies store small pieces of information in the user's browser.

Example: When you log in to YouTube and close the browser, cookies remember your login session. The next time you visit YouTube, you remain signed in and receive personalized video recommendations.

8.Flow Diagram



9.Conclusion

Application-layer protocols are essential for the functioning of the Internet. Every time users browse websites, watch videos, send emails, or use social media, protocols such as HTTP, DNS, FTP, and SMTP enable communication between devices and servers. DNS allows users to access websites using easy-to-remember names instead of IP addresses. HTTP transfers webpages and multimedia content efficiently, while TCP ensures reliable and error-free delivery of data. Cookies further improve the user experience by remembering preferences and login information. Without application-layer protocols, modern Internet services would not

be convenient, reliable, or scalable. They provide the foundation for communication, making every day online activities fast, secure, and accessible to millions of users worldwide.